

ASS. 4

1) (X, d) metric space.

Prove that d is a continuous fn on $X \times X$.

Hint since $d: X \times X \rightarrow \mathbb{R}$

Take two sequences $\{x_n\}, \{y_n\} \subset X$ s.t.

$x_n \rightarrow x \in X$ and $y_n \rightarrow y \in M$.

R.T.P.

$$d(x_n, y_n) \xrightarrow{n \rightarrow \infty} d(x, y).$$

In $\mathbb{R}$

2) Let $(X, d), (Y, \rho)$ be two metric spaces and A is a compact subset of X .

If $f: X \rightarrow Y$ is continuous, then $f(A)$ is compact. (sic) *continuity preserves compactness.*

3) Let E be a subset of a metric space (X, d) define a fn $f: X \rightarrow \mathbb{R}$ by

$$f(x) = d(x, E) = \inf \{d(x, y) : y \in E\}.$$

prove that 1) $f(x) = 0$ iff $x \in \overline{E}$.

more over f is ²⁾ continuous on X .
Moreover f is uniformly continuous on X (depends on ϵ).